Category Theory Notes

Yiwen Ding
yiwen.ding@example.com
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1 Category Theory

1.1 Definition of a Category

A **category** consists of:

- **Objects**
- **Morphisms** with domain and codomain
- **Composition** of morphisms

1.2 The Axioms for a Category

1.2.1 Associativity

For composable morphisms f, g, h:

$$(h \circ g) \circ f = h \circ (g \circ f)$$

$$(h \circ g) \circ f = h \circ (g \circ f)$$

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[rr, bend left, "g \circ f"] \arrow[rrr, bend left=60, "h \circ g"] & Y \arrow[r, "g"] \arrow[rr, bend left, "h"] & Z \arrow[r, "h"] & W \end{tikzcd}$$

1.2.2 Identity

For each object X, there exists an identity morphism $1_X : X \to X$ such that for any $f : X \to Y$:

$$1_X \circ f = f \quad \text{and} \quad f \circ 1_Y = f$$

$$1_X \circlearrowright X \xrightarrow{\quad f \quad} Y \circlearrowleft 1_Y$$

Question: Why is 1_X unique?

Proof: Assume there exist 1_X and $1'_X$ such that for all $f : X \to Y$ we have $1_X \circ f = f$ and $f \circ 1'_X = f$. Then:

$$1_X \circ 1'_X = 1'_X \quad \text{(taking } f = 1'_X \text{ in the first identity)}$$

$$1_X \circ 1'_X = 1_X \quad \text{(taking } f = 1_X \text{ in the second identity)}$$

Hence $1_X = 1'_X$. This concludes the proof.

Note: In such a category, any arrows can be composed. The set $\hom(X, X)$ with composition and 1_X forms a **monoid**.

1.3 Example: The Category Set

In the category **Set**, objects are sets A, B, C, D and morphisms are functions:

$$A \xrightarrow{f} C, \quad B \xrightarrow{g} D, \quad f \circ f' = f'', \quad g \circ g' = g''$$

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "f'"'] & C \arrow[d, "f''"] \\ A \arrow[r, "f"] & C \end{tikzcd} \qquad \begin{tikzcd} B \arrow[r, "g"] \arrow[d, "g'"'] & D \arrow[d, "g''"] \\ B \arrow[r, "g"] & D \end{tikzcd}$$

1.4 Product of Categories

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. The **product category** $\mathcal{C} \times \mathcal{D}$ has:

- Objects: pairs (A, B) with $A \in \mathcal{C}$, $B \in \mathcal{D}$
- Morphisms: $(f, g) : (A, B) \to (A', B')$ where $f : A \to A'$ in $\mathcal{C}$ and $g : B \to B'$ in $\mathcal{D}$

Composition: $(h, h') \circ (g, g') \circ (f, f') = (h \circ g \circ f, h' \circ g' \circ f')$

$$\begin{tikzcd} (A, B) \arrow[r, "(f,g)"] \arrow[d, "(f',g')"'] & (A', B') \arrow[d, "(f'',g'')"] \\ (A, B) \arrow[r, "(f,g)"] & (A', B') \end{tikzcd}$$

Proof of associativity in product:

$$(h \circ g) \circ f = h \circ (g \circ f), \quad (h' \circ g') \circ f' = h' \circ (g' \circ f')$$

Hence $(h, h') \circ ((g, g') \circ (f, f')) = ((h, h') \circ (g, g')) \circ (f, f')$. This concludes the proof.

1.5 Product Object with Projections

For objects A and B in a category, a **product** $A \times B$ with projections $\pi_1 : A \times B \to A$ and $\pi_2 : A \times B \to B$ satisfies the universal property:

$$\begin{tikzcd}
 & C \arrow[dl, "f"'] \arrow[d, dashed, "\exists!\, h"] \arrow[dr, "g"] & \\
A & A \times B \arrow[l, "\pi_1"] \arrow[r, "\pi_2"'] & B
\end{tikzcd}$$

For any object C with morphisms $f : C \to A$ and $g : C \to B$, there exists a unique $h : C \to A \times B$ such that $\pi_1 \circ h = f$ and $\pi_2 \circ h = g$.

1.6 Module / Exact Sequence Diagram

$$R^m \xrightarrow{\alpha} R^n \xrightarrow{\beta} N \longrightarrow 0$$

$$R^m \xrightarrow{\alpha} R^n \longrightarrow M \longrightarrow 0$$

Or as a short exact sequence:

$$0 \longrightarrow R^m \xrightarrow{\alpha} R^n \xrightarrow{\beta} M \longrightarrow 0$$

A Original Note (All Diagrams Preserved)

The following pages are the original PDF note, included to preserve all diagrams exactly as in the source.

Category Theory

A category consists of

(1) objects

(2) morphisms (domain and codomain)

(3) Composition of morphisms

The axioms for a category are:

(1) Associativity:

$$\begin{tikzcd}
\bullet \arrow[r, "f"'] \arrow[rr, bend left, "g\circ f"] \arrow[rrr, bend left=60, "h\circ(g\circ f)"] \arrow[rrr, bend right=60, "(h\circ g)\circ f"'] & \bullet \arrow[r, "g"'] \arrow[rr, bend right, "h\circ g"'] & \bullet \arrow[r, "h"'] & \bullet
\end{tikzcd}$$

(2) identity

$$\begin{tikzcd}
W \arrow[r, "f"] & X \arrow[r, "g"] \arrow[loop above, "1_X"] & Y
\end{tikzcd}$$

① $g\circ 1_X = g$

② $1_X \circ f = f$

$$\begin{tikzcd}
W \arrow[r, "f"] \arrow[rr, bend left, "1_X\circ f = f"'] \arrow[rrr, bend left=60, "(g\circ 1_X)\circ f = g\circ f"] \arrow[rrrr, bend right=60] & X \arrow[r, "1_X"] \arrow[rr, bend left, "g\circ 1_X = g"] & X \arrow[r, "g"] & Y
\end{tikzcd}$$

$$g \circ (1_X \circ f) = g \circ f$$

Question: why 1_X is unique?

proof: assume exist $1'_X$ s.t .....

$$\begin{tikzcd} \cdot \arrow[loop above, "1_X"] \arrow[loop below, "1_{X'}"] \end{tikzcd}$$

So $1_X = 1_X \circ 1_{X'} = 1_{X'}$

this concludes the proof.

Note:

$$\begin{tikzcd} X \arrow[loop above, "1_X"] \arrow[r, bend left, "g"] \arrow[rr, bend left=60, "f"] & \cdot \arrow[r] & \cdot \end{tikzcd}$$

any arrows in such category can be composed.

$(\hom(X,X), \circ, 1_X)$ is a monoid!

Set^2

$$\begin{tikzcd} A \arrow[r, "f"] & C \arrow[r, "f'"] & E \\ B \arrow[r, "g"'] & D \arrow[r, "g'"'] & F \end{tikzcd}$$

$$\begin{tikzcd} \langle A, B\rangle \arrow[r, "\langle f, g\rangle"] \arrow[dr, "\langle f', g'\rangle \circ \langle f, g\rangle = \langle f' \circ f, g' \circ g\rangle"'] & \langle C, D\rangle \arrow[d, "\langle f', g'\rangle"] \\ & \langle E, F\rangle \end{tikzcd}$$

Let C, D be categories

Product Category

$$(h\circ g)\circ f = h\circ(g\circ f)$$

$$\begin{tikzcd}
a \arrow[r, "f"] \arrow[rr, bend left, "g\circ f"] \arrow[rrr, bend left=60, "h\circ g\circ f"] & b \arrow[r, "g"] \arrow[rr, bend left, "h\circ g"] & c \arrow[r, "h"] & d \\
a' \arrow[r, "f'"'] \arrow[rr, bend right, "g'\circ f'"'] \arrow[rrr, bend right=60, "h'\circ g'\circ f'"'] & b' \arrow[r, "g'"'] \arrow[rr, bend right, "h'\circ g'"'] & c' \arrow[r, "h'"'] & d'
\end{tikzcd}$$

$$(h'\circ g')\circ f' = h'\circ(g'\circ f')$$

$$\begin{aligned}
&\langle h,h'\rangle\circ(\langle g,g'\rangle\circ\langle f,f'\rangle) \\
&= \langle h,h'\rangle\circ\langle g\circ f, g'\circ f'\rangle \\
&= \langle h\circ(g\circ f), h'\circ(g'\circ f')\rangle \\
&= \langle (h\circ g)\circ f, (h'\circ g')\circ f'\rangle \\
&= \langle h\circ g, h'\circ g'\rangle\circ\langle f,f'\rangle \\
&= (\langle h,h'\rangle\circ\langle g,g'\rangle)\circ\langle f,f'\rangle
\end{aligned}$$

This concludes the proof

$\mathbf{Set}^{\to}$

$$\begin{tikzcd}
A \arrow[r, "h"] \arrow[d, "f"'] & C \arrow[d, "g"] \\
B \arrow[r, "k"'] \arrow[ur, phantom, "\ulcorner"] & D
\end{tikzcd}
\qquad
\begin{tikzcd}
f \arrow[r, "\langle h,k\rangle"] \arrow[dr, "\langle j\circ h, l\circ k\rangle"'] & g \arrow[d, "\langle j,l\rangle"] \\
 & i
\end{tikzcd}$$

$$h\circ(l\circ j) = (h\circ l)\circ g$$

$$\begin{tikzcd}
a \arrow[r, "j"] \arrow[d, "i"'] \arrow[rr, bend left, "l\circ j"] \arrow[rrr, bend left=60, "h\circ l\circ j"] & b \arrow[r, "l"] \arrow[d, "q"] \arrow[rr, bend left, "h\circ l"] & c \arrow[r, "h"] \arrow[d, "r"] & d \arrow[d, "p"] \\
e \arrow[r, "k"'] \arrow[rr, bend right, "m\circ k"'] \arrow[rrr, bend right=60, "o\circ m\circ k"'] & f \arrow[r, "m"'] \arrow[rr, bend right, "o\circ m"'] & g \arrow[r, "o"'] & h
\end{tikzcd}$$

$$(o\circ m)\circ k = o\circ(m\circ k)$$

$$\begin{tikzcd}
A \arrow[r, "h"] \arrow[d, "f"'] \arrow[rr, bend left, "g\circ h"] & B \arrow[r, "g"] \arrow[d, "k"] & C \arrow[d, "l"] \\
D \arrow[r, "e"'] \arrow[rr, bend right, "j\circ e"'] & E \arrow[r, "j"'] & F
\end{tikzcd}$$

$$l\circ(g\circ h) = (j\circ e)\circ f \quad ?$$

$$\begin{array}{ccc}
\| & & \| \\
(l\circ g)\circ h & & j\circ(e\circ f) \\
\| & & \| \\
(j\circ k)\circ h & = & j\circ(k\circ h)
\end{array}$$

N, M

$D^{(0)}[S_m/x]$

$R_{\square}[\{n/a\}]$

$R_{\square}^{(1)}[\{n/a\}]$

$R_{\square}^{(0)}[\{n/a\}]$

$R_{\square}^{(1)}[\{m/x\}]$

$$[R_{\square}](c) = \left(R_{\square}^{(0)}[(1\,c)],\ \text{———}\uparrow\right)$$

$$\left(\text{———}\downarrow,\ R_{\square}^{(0)}[[c]]\right)$$